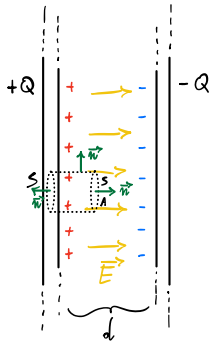
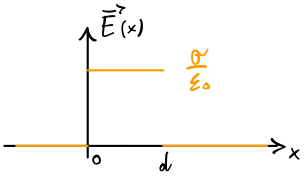
Condensatore Piano

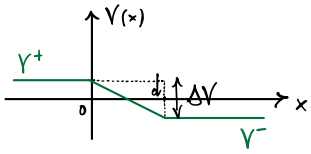


Induzione totale

$$\Phi_S(\vec{E}) = \int_A \vec{E}\cdot\vec{n}\,dS = E\int_A dS = EA \overset{\text{Gauss}}{=} \frac{\overbrace{\sigma A}^{q}}{\varepsilon_0}$$

$$\Rightarrow \vec{E} = \frac{\sigma}{\varepsilon_0}\vec{u}_x$$

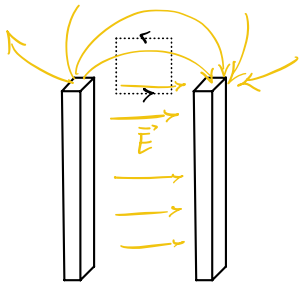




$$\int_{V(0)}^{V(d)} dV = -\int_0^d \vec{E}\cdot d\vec{l} = -\int_0^d E\,dx = -\int_0^d \frac{\sigma}{\varepsilon_0}\,dx = -\frac{\sigma}{\varepsilon_0}d \quad\Rightarrow\quad \boxed{\Delta V = \frac{\sigma}{\varepsilon_0}d}$$

$$C = \frac{Q}{\Delta V} = \frac{Q}{\frac{\sigma d}{\varepsilon_0}} = \frac{\sigma S}{\frac{\sigma d}{\varepsilon_0}} = \boxed{\frac{S\varepsilon_0}{d} = C}$$

055. Per pareti non infinite:

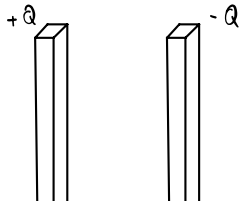


$\Gamma_\gamma(\vec{E}) \neq 0$ → non può essere che $\vec{E}$ è conservativo

$\Rightarrow \Gamma_\gamma(\vec{E}) = 0 \Rightarrow$ Effetti di Bordo

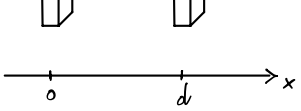
$\sqrt{\text{parete}} \gg d \Rightarrow$ Effetti di bordo trascurabili

Energia Elettrostatica di un Condensatore



$$U = U^+ + U^- \qquad (\text{considerando } U = \tfrac{1}{2}\int V\,dq = \tfrac{1}{2}VQ)$$

$$= \tfrac{1}{2}V^+Q^+ + \tfrac{1}{2}V^-Q^- = \tfrac{1}{2}V^+Q - \tfrac{1}{2}V^-Q = \tfrac{1}{2}Q(V^+ - V^-) = \tfrac{1}{2}Q\Delta V$$

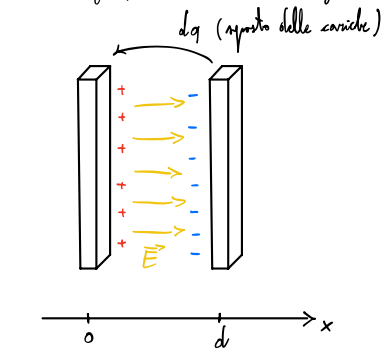


$|Q^+| = |Q^-|$

$\Rightarrow \boxed{U = \frac{1}{2} Q \Delta V} \longrightarrow = \frac{1}{2} Q \Delta V^2$

$\longrightarrow = \frac{1}{2} \frac{Q^2}{C}$

Cosa significa la nostra energia?



$dE_K = 0$ (le cariche, alla fine, sono ferme)

$dE_K = d\mathcal{L}_{ext} + d\mathcal{L}_{el} = 0 \qquad \Rightarrow d\mathcal{L}_{ext} = -d\mathcal{L}_{el} = dV$

$$d\mathcal{L}_{el} = -dU = \int_{dq}^{0} dq\, \vec{E}\cdot d\vec{l} = -dq \int_{V(d)=V^-}^{V(0)=V^+} dV = -dq\,\Delta V \qquad \Rightarrow \boxed{d\mathcal{L}_{el} = -dq\,\Delta V}$$

($d\vec{l}$ e $-dV$)

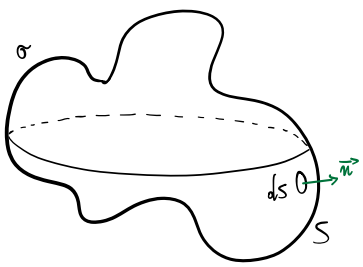
$$\mathcal{L}_{el} = \int d\mathcal{L}_{el} = \int -dq\,\Delta V = \int_0^Q -dq\,\frac{Q}{C} = -\frac{1}{2}\frac{Q^2}{C} = -U \quad \Rightarrow U = \frac{1}{2}\frac{Q^2}{C}$$

risultati già ottenuti!

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}\frac{(\sigma S)^2}{\frac{\varepsilon_0 S}{d}} = \frac{1}{2}\frac{\sigma^2}{\varepsilon_0} S d = \frac{1}{2}\varepsilon_0 E^2 \tau = u\,\tau = U$$

densità di campo · Volume

PRESSIONE ELETTROSTATICA

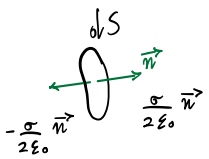


$\vec{E} = \frac{\sigma}{\varepsilon_0}\vec{n} \qquad S' = S - dS$

$\vec{E} = \vec{E}^{dS} + \vec{E}^{S'} \Rightarrow \vec{E}^{S'} = \vec{E} - \vec{E}^{dS} =$

fuori: $\frac{\sigma}{\varepsilon_0}\vec{n} - \frac{\sigma}{2\varepsilon_0}\vec{n} = \frac{\sigma}{2\varepsilon_0}\vec{n}$

dentro: $0 - \left(-\frac{\sigma}{2\varepsilon_0}\vec{n}\right) = \frac{\sigma}{2\varepsilon_0}\vec{n}$



$\Rightarrow \vec{F}\,dS = dq\,\vec{E}^{S'} = \sigma\, dS\, \frac{\sigma}{2\varepsilon_0}\vec{n} = \frac{\sigma^2}{2\varepsilon_0} dS\,\vec{n}$

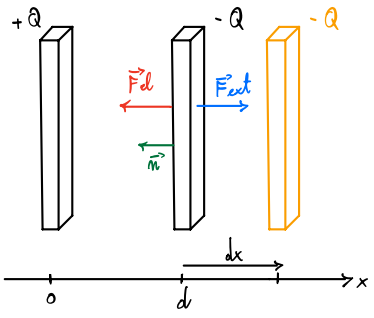
$$P = \frac{\vec{F}\,dS\cdot\vec{n}}{dS} = \frac{\sigma^2}{2\varepsilon_0}\frac{dS}{dS} = \frac{\sigma^2}{2\varepsilon_0} = \frac{1}{2}\varepsilon_0 E^2 = u \qquad \left[\frac{N}{m^2}\right] = [Pa] \qquad \Rightarrow \boxed{P = u}$$

Pressione elettrostatica

$\Rightarrow$ permette alle cariche di rimanere in superficie

subito fuori la superficie!

Condensatore Piano (Lavori Virtuali)



isolato $\Longrightarrow$ $Q = cost$

Svolgo un lavoro virtuale, sposto di dx un'armatura

$dE_K = 0$ $\quad dL_{el} + dL_{ext} = 0$ $\quad \Rightarrow dL_{ext} = -dL_{el}$

$$dL_{ext} = \vec{F}_{ext}\, d\vec{x} = F_x^{ext}\, dx$$
$$dL_{el} = \vec{F}_{el} \cdot d\vec{x} = F_x^{el}\, dx$$
$$\Big\} \; F_x^{el} = -F_x^{ext}$$

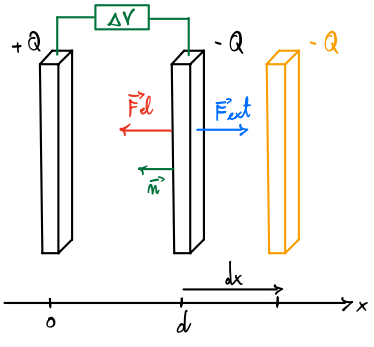
$dU = u_f \tau_f - u_i \tau_i = u(\tau_f - \tau_i) = u S\, dx$ per il campo non dipende dalla distanza fra le armature

$$dL_{el} = -dU = -u S\, dx$$
$$dL_{el} = F_x^{el}\, dx = -dU$$
$$\Big\} \; F_x^{el} = -\frac{dU}{dx} = -uS = -\frac{1}{2}\varepsilon_0 E^2 S = -\frac{1}{2}\frac{\sigma^2}{\varepsilon_0} S = -\frac{1}{2}\frac{Q^2}{\varepsilon_0 S}$$

$$\Rightarrow \boxed{\vec{F}_{el} = -\frac{1}{2}\frac{Q^2}{\varepsilon_0 S}\vec{u}_x}$$

$$\Rightarrow P = \frac{|\vec{F}_{el} \cdot \vec{n}|}{S} = \frac{F_x^{el}(\vec{u}_x)(-\vec{u}_x)}{S} = -\frac{F_x^{el}}{S} = \frac{uS}{S} = u$$

Generatore di Tensione



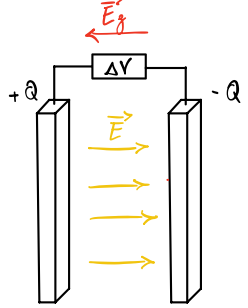
$\Rightarrow$ oggetto che mantiene $\Delta V = cost$.

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} C \Delta V^2 \quad \Rightarrow dU = \frac{1}{2}\Delta V^2\, dC \overset{(=)}{\underset{C = \frac{\varepsilon_0 S}{x}}{\downarrow}} \frac{1}{2}\Delta V^2 \left(-\frac{\varepsilon_0 S}{x^2}\, dx\right) < 0$$

$$F_x^{el} = -\frac{dU}{dx} = +\frac{1}{2}\Delta V^2 \frac{\varepsilon_0 S}{x^2} = \frac{1}{2}\frac{\Delta V^2}{\varepsilon_0 S}\underbrace{\frac{(\varepsilon_0 S)^2}{x^2}}_{C^2} = \frac{1}{2}\frac{\Delta V^2}{C^2}\frac{1}{\varepsilon_0 S} = \frac{1}{2}\frac{Q^2}{\varepsilon_0 S}$$

(dovrebbe avere il $-$, per il risultato precedente)

Perché?



oss. Quando allontano le armature, il campo diminuisce ($\Delta V = E\,d$), ma $E = \frac{\sigma}{\varepsilon_0}$, quindi anche σ diminuisce.

$U = u\,\tau = \frac{1}{2}\varepsilon_0 E^2 \tau$, dove $E^2 \propto \frac{1}{d^2}$ e $\tau \propto dx$, complessivamente $U \propto \frac{1}{dx}$

$\Rightarrow$ U diminuisce allontanando le armature.

Chi fa diminuire l'energia? Il generatore "mangia" l'energia.

$dE_K = 0 \quad \Rightarrow \quad dL^{el} + dL^{ext} + dL^{g} = dE_K = 0$

$dL^{ext} = -dL^{el} - dL^{g} = dU + dU_g$, dove sappiamo $dU = \frac{1}{2}\Delta V^2\, dC < 0$

$dL^{g} = \int dq\, \underbrace{\vec{E}_g \cdot d\vec{l}}_{-dV} = -\int dq\, dV = -dq \int_{V^+}^{V^-} dV = dq \int_{V^-}^{V^+} dV = \boxed{dq\,\Delta V = dL^{g}}$

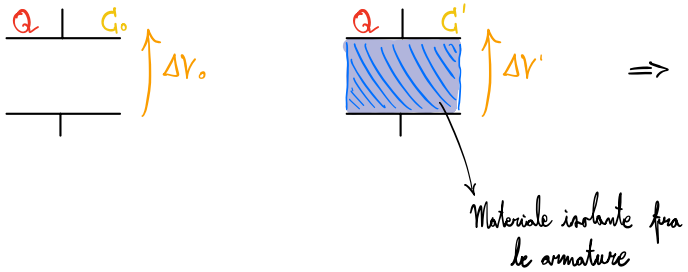
$\Rightarrow dU_g = -dL^{g} = -dq\,\Delta V = -\Delta V\, dC\, \Delta V = -dC\,\Delta V^2 > 0$

$\Rightarrow dL^{ext} = dU + dU_g = \frac{1}{2}\Delta V^2 dC - \Delta V^2 dC = -\frac{1}{2}\Delta V^2 dC > 0 \quad \Rightarrow \boxed{dU = -\frac{1}{2}\Delta V^2 dC}$

$dL^{ext} = F_x^{ext}\, dx = -F_x^{el}\, dx = -dU$

$\Rightarrow \boxed{F_x^{el} = \frac{dU}{dx}}$

DIELETTRICI (Materiali isolanti)



$\Rightarrow \Delta V' < \Delta V_0 \qquad C' > C_0$

$\boxed{\varepsilon_r := \frac{C}{C_0}}$ Costante dielettrica relativa

C_0 = capacità nel vuoto, C = capacità con il dielettrico, $\varepsilon_r > 1$ sperimentalmente

$\varepsilon = \varepsilon_0 \cdot \varepsilon_r$ Costante dielettrica assoluta